

RAL-MoE-ROM: Hierarchical Mixture-of-Experts for Regime-Adaptive Reduced Dynamics

Anonymous authors
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Abstract

Learning reduced dynamical models over broad parameter ranges is challenging when the underlying system exhibits qualitatively different regimes. A single global representation must simultaneously encode distinct local state geometries, while a single learned dynamical model must reproduce different stability, growth, and saturation mechanisms under recursive rollout. We introduce RAL-MoE-ROM, a two-level hierarchical mixture-of-experts framework for regime-adaptive reduced dynamics. The parameterized solution set is represented by overlapping regime-local reduced charts, each equipped with an autonomous physics-data specialist. Within each specialist, a structured sparse mixture of experts learns corrections to the projected reduced dynamics using nonlinear, affine, and low-rank bilinear expert branches selected by hierarchical sparse routing. A second routing level operates across regime-local specialists. Because the local charts have independent reduced coordinates, adjacent specialists are advanced independently and combined only after reconstruction in the common physical space through a learned Top-2 convex correction. Across three parametrized incompressible-flow benchmarks spanning steady, Hopf-onset, and periodic dynamics, the proposed specialists provide accurate long-horizon autonomous predictions, while the hierarchical fusion consistently improves predictions in evaluated transition regions. Ablations further show complementary contributions from the structured expert parameterization, the projected physical backbone, regime localization, and learned cross-regime routing.

1 Introduction

Learning compact dynamical models is central to scientific machine learning when high-fidelity simulation is too expensive for repeated evaluation. The problem is especially difficult when a physical parameter changes the qualitative dynamics, not only the response magnitude. Parameterized incompressible flows are representative: different parts of the parameter domain may relax to a steady state, exhibit weak oscillatory growth near instability onset, or sustain nonlinear periodic motion. A many-query reduced model must remain accurate under autonomous rollout across all of these regimes. Unlike a snapshot interpolant, it must continue evolving after the short initial history is exhausted, without injecting reference states into the prediction window.

Broad parameter ranges create two coupled difficulties. First, the state representation is regime dependent: a basis that efficiently describes nearly steady states need not compactly represent developed periodic dynamics. Second, the reduced evolution law must reproduce distinct mechanisms such as decay, growth, phase evolution, and saturation. A global model must solve both problems simultaneously, while recursive prediction continually feeds its own errors back into the dynamics. The difficulty is therefore not only model capacity: global coordinates can spend dimensions reconciling distinct state geometries, and one global evolution law can average mechanisms that should remain regime specific.

Projection-based reduced-order models provide explicit low-dimensional structure through POD and Galerkin projection (Sirovich, 1987; Berkooz et al., 1993; Benner et al., 2015),

but truncated interactions can degrade closure and long-horizon behavior (Amsallem & Farhat, 2012). Learned dynamics offer richer nonlinear maps, yet one-step accuracy need not survive recursive use (Pathak et al., 2018; Vlachas et al., 2018; Duraisamy et al., 2019; Brunton et al., 2020). Hybrid methods therefore combine a projected backbone with learned unresolved terms (Dar et al., 2023; Lee & Carlberg, 2020; Fresca & Manzoni, 2022; Geelen et al., 2023). A complementary approach replaces one global representation with local subspaces, clusters, or manifolds (Kaiser et al., 2014; Li et al., 2021; Diaz et al., 2024). These two remedies address complementary parts of the problem—the first improves the dynamical inductive bias, while the second reduces representation mismatch.

Localization, however, creates an assembly problem. Independently constructed charts have different affine means, bases, scalings, and reduced operators, so their coefficient vectors have no canonical componentwise correspondence. Switching or averaging those vectors can therefore mix incompatible representations. Mixture-of-experts (MoE) models provide a natural language for conditional computation (Jacobs et al., 1991; Jordan & Jacobs, 1994; Shazeer et al., 2017; Fedus et al., 2022), but conventional MoE layers typically route predictors acting in a shared feature space. Regime-dependent reduced dynamics instead require both conditional closure within a chart and selection or combination of complete autonomous models across noncommensurate charts. The outer decision must also respect which local models are supported near a regime boundary; router confidence alone does not make a cross-chart combination dynamically meaningful.

We introduce the Regime-Adaptive Localized Mixture-of-Experts Reduced-Order Model (RAL-MoE-ROM), a two-level hierarchy for this setting. Overlapping regime-local charts each support an autonomous physics–data specialist. Within a specialist, hierarchical sparse routing selects structured correction experts with nonlinear, affine, and low-rank bilinear branches while retaining the projected dynamics as an explicit inductive bias. Group Top-1 and channel Top-2 decisions activate a shared expert and a small routed subset for the velocity and pressure channels. Specialists are trained with closed-loop multi-step objectives so that the learned correction is exposed to its own recursive inputs. Across specialists, a parameter-only E2 router produces one trajectory-level regime preference. Physical-history descriptors are used only by the Top-2 convex correction (T2-C) in an admissible overlap. The two selected specialists encode the same physical history in their native charts, evolve independently, and are combined only after reconstruction on the common physical mesh. The fused field is an output and is never fed back into either local rollout.

Our contributions are threefold:

- Two-level hierarchical routing. Sparse expert routing models unresolved dynamics inside each local specialist, while a second level selects or combines complete specialists across regimes. This distinguishes conditional closure from conditional selection of an entire dynamical system.
- Structured closures for autonomous reduced dynamics. Nonlinear, affine, and low-rank bilinear expert branches augment an explicit projected backbone and are trained under closed-loop rollout. Separate velocity and pressure routing allows the two algebraically different correction channels to adapt independently.
- Physical-space assembly of noncommensurate local models. Adjacent specialists retain independent coordinates and rollouts; T2-C couples only their reconstructed fields, avoiding coefficient or operator interpolation. A learned history-conditioned correction refines the parameter-only routing preference while preserving a convex output.

We evaluate RAL-MoE-ROM on three parameterized incompressible-flow systems spanning steady, Hopf-onset, and periodic dynamics. The experiments separate the effects of the structured expert map, projected backbone, localization, and cross-regime assembly. Over the evaluated finite horizons, the local specialists remain accurate and finite, and T2-C improves the corresponding hard or single-specialist prediction in the tested overlaps. The comparisons include vanilla experts, data-only dynamics, a global chart, POD–Galerkin, one-step POD–DeepONet, hard routing, and fixed or probability-based blends. Localization

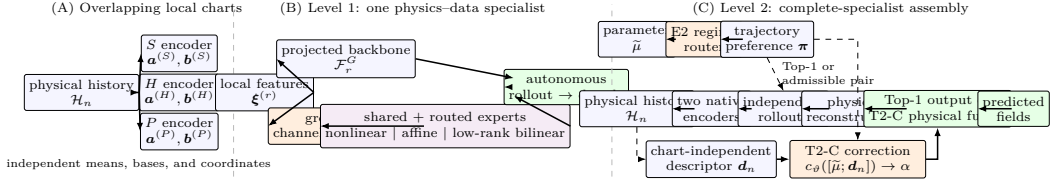


Figure 1: Overview of RAL-MoE-ROM. Sparse routing acts within each regime-local specialist, while parameter-only regime routing selects complete local models. Physical-history descriptors condition only T2-C, which combines independently evolved specialists after reconstruction in the common physical space.

is not uniformly superior to a global model at every parameter; its value emerges from combining local coordinates, regime-specific dynamics, and adaptive assembly.

2 Related Work

Projection, learned dynamics, and autonomous rollout. POD–Galerkin methods project governing equations onto a data-adapted basis and retain an interpretable reduced vector field (Sirovich, 1987; Berkooz et al., 1993; Benner et al., 2015). Their long-horizon accuracy can be limited by truncated scales, imperfect closure, and reduced model stabilization (Amsallem & Farhat, 2012). Non-intrusive models instead learn reduced or latent dynamics (Hesthaven & Ubbiali, 2018; Lee & Carlberg, 2020; Champion et al., 2019; Park et al., 2024), and autoregressive forecasters demonstrate the expressivity of learned evolution laws for complex temporal systems (Pathak et al., 2018; Vlachas et al., 2018). Neural operators learn maps between function spaces (Lu et al., 2021; Kovachki et al., 2023), whereas hybrid ROMs retain physical or projection structure while learning unresolved components (Duraissamy et al., 2019; Brunton et al., 2020; Dar et al., 2023; Fresca & Manzoni, 2022; Geelen et al., 2023). RAL-MoE-ROM follows the hybrid view, but replaces a single residual map with a conditionally sparse correction trained under closed-loop rollout. This distinction is important because low one-step error alone does not establish reliable recursive prediction.

Localized representations and conditional dynamics. Local reduced bases, operator localization, clustering, and nonlinear local manifolds reduce the representational burden imposed on a single global chart (Amsallem & Farhat, 2008; Peherstorfer et al., 2014; Kaiser et al., 2014; Li et al., 2021; Diaz et al., 2024). These approaches motivate parameter-local representations, but localization also makes the dynamics chart dependent: independently constructed models generally differ in their means, bases, scaling, and operators. RAL-MoE-ROM treats each chart as a complete autonomous specialist and does not assume that two local coefficient vectors are commensurate. Its cross-regime problem is therefore not merely basis selection; it is conditional assembly of independently evolved dynamical systems.

Mixture-of-experts and the cross-chart gap. Classical hierarchical mixtures learn conditional expert assignments (Jacobs et al., 1991; Jordan & Jacobs, 1994), and sparse MoE architectures scale conditional computation by activating only a subset of experts (Shazeer et al., 2017; Fedus et al., 2022). In most such models, the routed predictors consume and produce a shared representation. RAL-MoE-ROM uses two different routing levels. Inside a local chart, group and channel routers activate structured closure experts that share the same reduced coordinates. Outside the chart, the E2 router selects complete autonomous specialists whose coordinates are unrelated. An admissible adjacent pair is never averaged in coefficient space: both models roll out independently and T2-C fuses only their reconstructed physical fields. This separation of within-chart conditional closure from cross-chart dynamical assembly is the central gap addressed here.

3 RAL-MoE-ROM

We consider parameterized dynamics whose trajectories occupy qualitatively different regions of state space as the physical parameter μ varies. RAL-MoE-ROM separates representation, within-regime closure, and cross-regime assembly instead of forcing every trajectory into one latent coordinate system.

3.1 Overlapping regime-local reduced charts

Let $r \in \{S, H, P\}$ index steady, Hopf-onset, and periodic regimes, and let $\mathcal{M}_r$ denote overlapping parameter regions. Each chart has its own affine mean, velocity and pressure bases, scaling, and reduced operators. A physical state is encoded and reconstructed as

$$\mathbf{a}^{(r)} = \Phi_{u,r}^\top \mathbf{W}_u (\mathbf{u} - \bar{\mathbf{u}}_r), \quad \hat{\mathbf{u}}^{(r)} = \bar{\mathbf{u}}_r + \Phi_{u,r} \mathbf{a}^{(r)}, \quad (1)$$

with the analogous map for gauge-corrected pressure coefficients $\mathbf{b}^{(r)}$. Because these coordinates are constructed independently, components of $\mathbf{a}^{(r)}$ and $\mathbf{a}^{(s)}$ are not assumed to correspond when $r \neq s$. This is the reason cross-regime fusion is deferred to physical space.

3.2 Physics-data specialists with structured sparse experts

Projection supplies a regime-local dynamical backbone,

$$\dot{\mathbf{a}}^{(r)} = \underbrace{\mathbf{c}_r + \mathbf{A}_r \mathbf{a}^{(r)} + \mathbf{H}_r : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + \mathbf{C}_r^p \mathbf{b}^{(r)}}_{\mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu)} + \rho_{\theta_r}^u(\boldsymbol{\xi}^{(r)}). \quad (2)$$

The feature vector $\boldsymbol{\xi}^{(r)}$ contains standardized local coordinates, the projected right-hand side, the parameter, and a short state history. The learned correction is a channel-wise sparse MoE,

$$\mathcal{M}_{\theta_r}^c(\boldsymbol{\xi}) = \omega_0^c E_{0,g^*}^c(\boldsymbol{\xi}) + \sum_{e \in \mathcal{T}^c} \omega_e^c E_{e,g^*}^c(\boldsymbol{\xi}), \quad E(\boldsymbol{\xi}) = N(\boldsymbol{\xi}) + W_{\text{lin}} \boldsymbol{\xi} + \kappa W_o [(U^\top \boldsymbol{\xi}) \odot (V^\top \boldsymbol{\xi})]. \quad (3)$$

The first routing decision selects a sparse expert group and the second selects experts within that group. The nonlinear, affine, and low-rank bilinear branches provide complementary correction structures without removing the explicit projected term. Pressure is reconstructed through a physics-data algebraic map rather than evolved as an unconstrained recurrent state; details are in Appendix C.

3.3 Closed-loop autonomous training

Each specialist is trained on rollout windows. Starting from encoded physical history, the numerical integrator repeatedly evaluates Eq. (2) on its own predictions. The loss aggregates velocity- and pressure-coordinate errors across the window, so the optimization sees the distribution shift produced by recursive use. All comparisons use the same held-out trajectories and the reported rollout horizons; exact splits are summarized in Table 1 and Appendix E.

3.4 Regime routing and physical-space T2-C fusion

The outer E2 router depends only on the standardized physical parameter $\tilde{\mu}$ and produces a trajectory-level regime preference $\boldsymbol{\pi}(\mu) = \text{softmax}(\mathcal{R}_{E2}(\tilde{\mu}))$. Its probabilities are evaluated once and remain fixed during the rollout. Chart-independent descriptors extracted from the initial physical history are used only by the T2-C correction gate in an admissible overlap; they are not E2 inputs. Hard Top-1 prediction advances the largest-probability admissible specialist. When an adjacent pair is admissible ($S-H$ or $H-P$), both specialists encode the same physical history in their native coordinates and then evolve independently. For reconstructed fields $\hat{\mathbf{q}}^{(r)}$ and $\hat{\mathbf{q}}^{(s)}$, T2-C predicts a window-conditioned convex weight and forms

$$\hat{\mathbf{q}}_{\text{T2C}}(t) = \alpha \hat{\mathbf{q}}^{(r)}(t) + (1 - \alpha) \hat{\mathbf{q}}^{(s)}(t), \quad \alpha = \sigma(\text{logit}(\bar{\pi}_r) + \Delta_\psi). \quad (4)$$

Table 1: Benchmark and held-out autonomous-rollout protocol. Entries list train/validation/test trajectories by local chart.

Benchmark	S chart	H chart	P chart
Circular cylinder	20.0–46.4; 14/2/4; $K=56$	45.5–59.2; 29/2/3; $K=56$	57.3–200; 53/6/4; $K=48$
Centered square	50–95.4; 60/5/4; $K=16$	94–102; 23/6/5; $K=48$	98.5–150; 27/6/4; $K=24$
Fluidic pinball	10–19; 37/8/7; $K=56$	16.6–25; 39/8/7; $K=56$	21.25–60; 47/9/9; $K=32$

Table 2: Fluidic-pinball autonomous prediction. Errors are percentages; “div.” counts divergent evaluation windows.

Regime (K)	Model	E_u	E_p	E_{joint}	div.
Steady (56)	POD–Galerkin	–	–	–	448/448
	RAL–MoE–ROM	0.2247	0.7593	0.9840	0
Hopf (56)	POD–Galerkin	0.4378	48.50	48.94	141/320
	RAL–MoE–ROM	0.0995	1.986	2.085	0
Periodic (32)	POD–Galerkin	1.641	41.56	43.20	0
	RAL–MoE–ROM	0.2897	0.5518	0.8415	0
Periodic core (56)	POD–Galerkin	2.555	39.36	41.91	0
	RAL–MoE–ROM	0.6222	1.050	1.672	0

Here $\bar{\pi}_r$ is the pair-normalized E2 probability and $\Delta_\psi = c_\psi([\tilde{\mu}; \mathbf{d}_n])$ is a learned correction based on the standardized parameter and chart-independent physical-history descriptor $\mathbf{d}_n$. Fusion occurs after reconstruction and is not fed back into either local rollout. Admissibility is selected from validation support and should not be interpreted as a formal stability certificate.

4 Experiments

We study three two-dimensional incompressible-flow systems with Reynolds number as the parameter: flow past a circular cylinder, flow past a centered square cylinder, and the fluidic pinball. Each dataset spans steady, Hopf-onset, and periodic behavior. Errors E_u and E_p are relative field errors in percent, averaged using the protocol of each benchmark; $E_{\text{joint}} = E_u + E_p$ where reported. “Divergent” denotes an evaluation window that violates the predefined rollout-divergence criterion used in the original experiment; this includes non-finite states and threshold violations. We preserve the numerical values and held-out partitions of the journal experiments.

4.1 Do local specialists support long autonomous rollouts?

On the circular cylinder, the proposed specialists remain finite across every evaluated regime. In the steady regime, their velocity-error range is 0.0330–0.1723%, compared with 0.0384–0.2995% for a vanilla learned specialist and 0.0845–0.3404% for the global model. In the Hopf regime the vanilla model is not evaluable, while RAL–MoE–ROM gives $E_u = 0.0032$ –0.0288% and $E_p = 0.0355$ –0.4321%. In the periodic regime it improves the velocity-error range over vanilla from 1.0393–3.0836% to 0.3809–1.3473%. Full pressure ranges and the data-only ablation are in Appendix F.

The centered-square specialists transfer the construction to a different geometry and mesh. Aggregate held-out errors are $(E_u, E_p) = (1.5311, 1.4583)\%$ for the steady chart, $(0.4387, 0.4844)\%$ for Hopf, and $(0.9395, 1.0921)\%$ for periodic flow. On fluidic pinball, the comparison in Table 2 shows that the learned specialist removes the divergent rollouts of the projected baseline in the steady and Hopf settings and substantially reduces joint error across all four evaluations. This is empirical long-horizon evidence under the tested distributions, not a guarantee of stability outside them.

Table 3: Centered-square routing and fusion at $K = 24$. Lower is better.

	Overlap	Predictor	E_u	E_p	E_{joint}
$S-H$		E2 Top-1	0.4942	1.2356	1.7298
		Equal blend	1.3644	1.6677	3.0321
		E2-probability blend	0.7898	1.3949	2.1847
		T2-C	0.4615	0.7343	1.1958
		Convex oracle	0.4659	0.7183	1.1842
$H-P$		E2 Top-1	0.7422	1.1052	1.8473
		Equal blend	0.6077	1.0069	1.6146
		E2-probability blend	0.6154	1.0464	1.6618
		T2-C	0.4509	0.3956	0.8465
		Convex oracle	0.4479	0.3905	0.8384

Table 4: Compact ablation summary using existing headline metrics. Regime columns report circular-cylinder E_u/E_p ranges; the overlap column reports centered-square $S-H/H-P$ joint errors at $K = 24$. A dash indicates that the comparison does not modify that component.

Variant	Steady E_u/E_p	Hopf E_u/E_p	Periodic E_u/E_p	Overlap E_{joint}
Full RAL-MoE-ROM	.0330–.1723 / .2500–1.3249	.0032–.0288 / .0355–.4321	.3809–1.3473 / 1.6825–5.9601	1.1958 / .8465
FNN experts	.0384–.2995 / 2.2163–18.5603	N/E	1.0393–3.0836 / 3.9659–13.3257	–
Data only	.1637–.5122 / 4751.1–27275.5	.1260–.3880 / 1280.3–1524.6	.7854–1.8677 / 2.2511–10.8189	–
Global chart	.0845–.3404 / 1.5226–7.4363	.0110–.0268 / .0560–.2882	.5873–1.1726 / 2.1896–6.3566	–
E2 Top-1 only	–	–	–	1.7298 / 1.8473
E2-probability blend	–	–	–	2.1847 / 1.6618

4.2 Does learned physical-space fusion improve overlap prediction?

Table 3 separates hard routing from field-space combination on the centered-square benchmark. T2-C improves E2 Top-1 in both overlaps and is close to the a posteriori window-wise convex oracle. Equal weights and direct E2-probability weights do not reproduce the gain, indicating that the improvement is not a generic consequence of averaging. At $Re = 95.1$, T2-C reduces joint error from 1.7140% to 1.1827%; at $Re = 100.5$, it reduces 2.6485% to 0.8571%. Figure 2 shows the corresponding physical fields and velocity-error maps. The complete parameter-wise table and fluidic-pinball overlap study appear in Appendices G and H.

4.3 What contributes to the result?

The original ablations isolate four effects. Removing the projected backbone produces the data-only degradation visible on circular-cylinder rollouts. Replacing the local charts with a global representation changes the regime-wise error profile and removes regime-specific coordinates, but does not uniformly degrade physical-field accuracy. This comparison indicates that localization is most useful when combined with regime-specific dynamics and cross-regime assembly rather than as a universally superior representation. Replacing structured sparse experts with the vanilla learned correction widens long-horizon errors and fails in the evaluated Hopf case; replacing T2-C with Top-1, equal, or raw-probability fusion worsens overlap prediction. Together these results support the architectural decomposition rather than a claim that any single component is uniformly dominant.

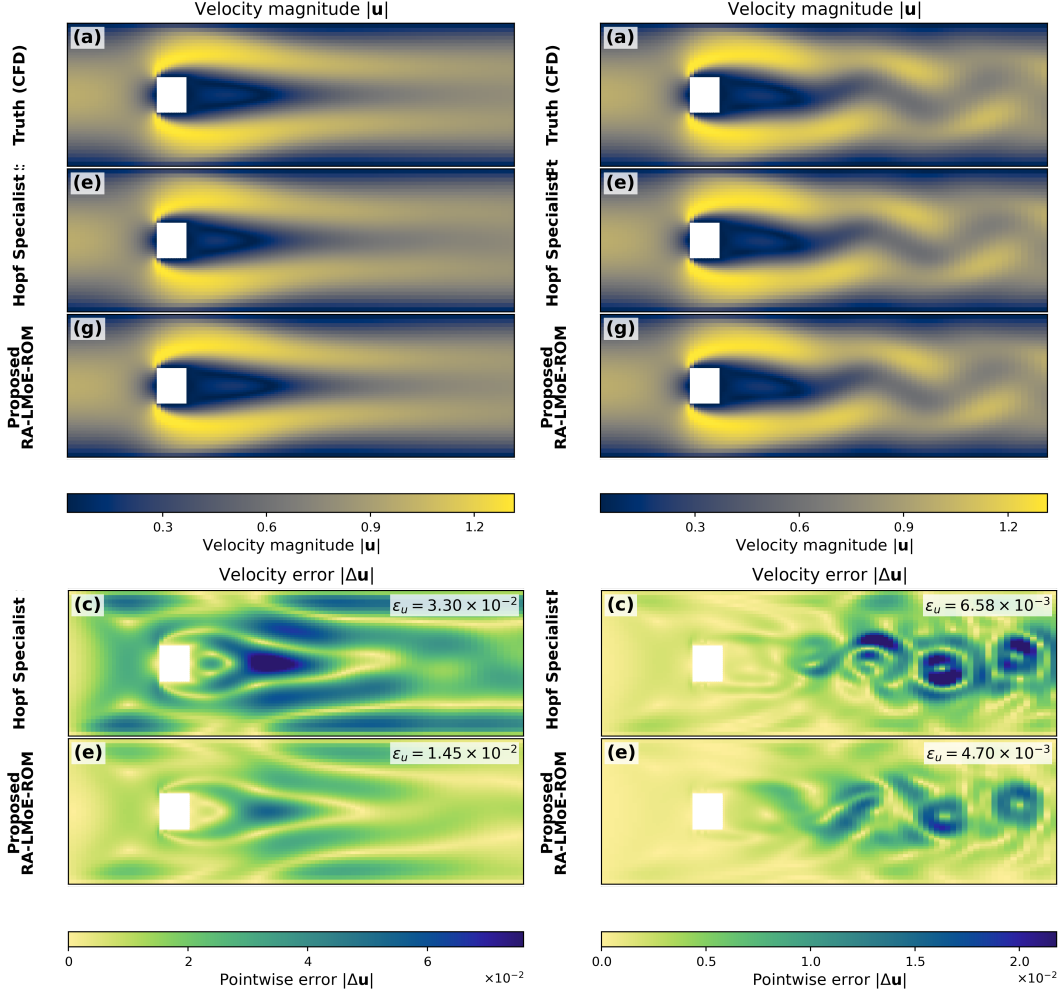


Figure 2: Representative cross-regime predictions on the centered-square benchmark. The left and right panels correspond to the $S-H$ and $H-P$ overlaps, respectively. T2-C combines independently evolved specialists after physical reconstruction and reduces the velocity-field discrepancy relative to the corresponding single-specialist prediction. Complete velocity and pressure fields are reported in Appendix G.

5 Limitations

The regime cover and admissible adjacency graph are selected using validation support; the method does not discover arbitrary bifurcation structure or certify unseen transitions. T2-C is evaluated only between adjacent specialists, and convex physical-space fusion may be insufficient when more than two regimes interact or when conserved quantities require a constrained assembly rule. The evidence concerns two-dimensional incompressible flows and empirical finite-horizon behavior; it does not establish provable stability, conservation, or accuracy outside the sampled parameter ranges. A separate hierarchy is trained for each benchmark, so cross-geometry transfer of one hierarchy is not evaluated. Multiple local bases, operators, and specialists also increase training and model-management cost relative to one global ROM.

The retained experiment package does not include the final training checkpoints and complete architecture configurations. We therefore do not report post-hoc internal expert-utilization or non-collapse statistics, nor a total-versus-active parameter table; the structured-expert comparison should not be interpreted as a parameter-matched capacity

test. The retained results are from fixed training runs, and multi-seed uncertainty has not been quantified. Finally, the source records define rollout divergence to include non-finite states and violations of a predefined threshold, but do not retain that threshold’s numerical value. These configuration, seed, and threshold records must be recovered for the final reproducibility package rather than inferred.

6 Conclusion

RAL-MoE-ROM casts broad-parameter reduced dynamics as a two-level mixture-of-experts problem. Structured sparse experts learn unresolved dynamics inside each physics–data specialist, while an outer router selects or combines complete local models. Keeping the local coordinates independent and coupling only reconstructed physical fields avoids interpolation between incompatible modal states. Across the three evaluated flow systems, the specialists achieve low error over the evaluated autonomous rollout horizons, and T2-C improves prediction in the tested transition regions. The results motivate extensions that learn the regime cover, impose physical constraints on field-space fusion, and evaluate the hierarchy on higher-dimensional and non-fluid dynamical systems.

Reproducibility statement

The main text reports the data splits, rollout horizons, reduced dimensions, baselines, and evaluation metrics needed to interpret the results. Appendix E documents the local-coordinate construction, specialist architecture, routing procedure, and additional benchmark results. We report only implementation details supported by the retained experiment sources; the remaining artifact checklist identifies material that must be assembled before submission.

AI use statement

TODO: authors to complete final disclosure.

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A Full problem setup and weighted POD construction

A.1 Parameterized full-order model, gauge, and regime cover

After spatial discretization, the velocity $\mathbf{u}(t; \mu) \in \mathbb{R}^{N_u}$ and pressure $p(t; \mu) \in \mathbb{R}^{N_p}$ satisfy

$$\dot{\mathbf{u}} = \mathbf{f}_h(\mu) + \mathcal{L}_h(\mu)\mathbf{u} - \mathcal{N}_h(\mathbf{u}, \mathbf{u}; \mu) - \nabla_h(\mu)p, \quad \text{div}_h(\mu)\mathbf{u} = 0. \quad (5)$$

Pressure is defined only up to an additive constant. With positive finite-volume weights $\mathbf{w}$, every pressure snapshot is mapped to the weighted zero-mean gauge

$$\Gamma_p(q) = q - \frac{\mathbf{w}^\top q}{\mathbf{w}^\top \mathbf{1}} \mathbf{1}, \quad p^\circ = \Gamma_p(p), \quad \mathbf{w}^\top p^\circ = 0. \quad (6)$$

The parameter domain is covered by overlapping regions

$$\mathcal{M} \subseteq \bigcup_{r \in \{S, H, P\}} \mathcal{M}_r, \quad \mathcal{O}_{SH} = \mathcal{M}_S \cap \mathcal{M}_H, \quad \mathcal{O}_{HP} = \mathcal{M}_H \cap \mathcal{M}_P. \quad (7)$$

No direct S – P overlap is used.

A.2 Area-weighted local POD

The finite-volume quadrature induces

$$\langle \mathbf{v}, \mathbf{z} \rangle_{M_u} = \mathbf{v}^\top M_u \mathbf{z}, \quad \langle q, s \rangle_{M_p} = q^\top M_p s, \quad M_p = \text{diag}(\mathbf{w}), \quad M_u = \text{diag}(M_p, M_p). \quad (8)$$

For chart r , let $X_u^{(r)}$ and $X_p^{(r)}$ contain velocity and gauge-corrected pressure snapshots after subtracting their chart-specific means $\bar{\mathbf{u}}^{(r)}$ and $\bar{p}^{(r), \circ}$. The local bases solve

$$\Phi_u^{(r)} \in \arg \min_{\Psi^\top M_u \Psi = I_{d_u}} \left\| X_u^{(r)} - \Psi \Psi^\top M_u X_u^{(r)} \right\|_{M_u, F}^2, \quad (9)$$

$$\Phi_p^{(r)} \in \arg \min_{\Psi^\top M_p \Psi = I_{d_p}} \left\| X_p^{(r)} - \Psi \Psi^\top M_p X_p^{(r)} \right\|_{M_p, F}^2, \quad (10)$$

where $\|Y\|_{M, F}^2 = \text{tr}(Y^\top M Y)$. Equivalently, for the weighted singular value decompositions

$$M_u^{1/2} X_u^{(r)} = U_u^{(r)} \Sigma_u^{(r)} V_u^{(r)\top}, \quad M_p^{1/2} X_p^{(r)} = U_p^{(r)} \Sigma_p^{(r)} V_p^{(r)\top}, \quad (11)$$

the bases are

$$\Phi_u^{(r)} = M_u^{-1/2} U_u^{(r)}(:, 1:d_u), \quad \Phi_p^{(r)} = M_p^{-1/2} U_p^{(r)}(:, 1:d_p), \quad (12)$$

and satisfy $\Phi_u^{(r)\top} M_u \Phi_u^{(r)} = I$ and $\Phi_p^{(r)\top} M_p \Phi_p^{(r)} = I$. Gauge consistency also gives $\mathbf{w}^\top \Phi_p^{(r)} = \mathbf{0}^\top$.

Encoding and decoding are chart specific:

$$\mathbf{a}^{(r)} = \Phi_u^{(r)\top} M_u (\mathbf{u} - \bar{\mathbf{u}}^{(r)}), \quad \mathcal{D}_r^u(\mathbf{a}^{(r)}) = \bar{\mathbf{u}}^{(r)} + \Phi_u^{(r)} \mathbf{a}^{(r)}, \quad (13)$$

$$\mathbf{b}^{(r)} = \Phi_p^{(r)\top} M_p (\Gamma_p(p) - \bar{p}^{(r),\circ}), \quad \mathcal{D}_r^p(\mathbf{b}^{(r)}) = \bar{p}^{(r),\circ} + \Phi_p^{(r)} \mathbf{b}^{(r)}. \quad (14)$$

Learned components use local standardizations

$$\tilde{\mathbf{a}}^{(r)} = (\mathbf{a}^{(r)} - \boldsymbol{\mu}_a^{(r)}) \oslash \boldsymbol{\sigma}_a^{(r)}, \quad \tilde{\mathbf{b}}^{(r)} = (\mathbf{b}^{(r)} - \boldsymbol{\mu}_b^{(r)}) \oslash \boldsymbol{\sigma}_b^{(r)}. \quad (15)$$

B Projected Galerkin and pressure operators

B.1 Chart-local momentum dynamics

Substituting the affine reconstructions into Eq. (5) and testing against the local velocity basis gives

$$\dot{\mathbf{a}}^{(r)} = \mathbf{c}_r(\mu) + A_r(\mu) \mathbf{a}^{(r)} + H_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + C_r^p(\mu) \mathbf{b}^{(r)}. \quad (16)$$

For velocity modes $\phi_i^{u,(r)}$ and pressure modes $\phi_\ell^{p,(r)}$,

$$[\mathbf{c}_r]_i = \left\langle \phi_i^{u,(r)}, \mathbf{f}_h + \mathcal{L}_h \bar{\mathbf{u}}^{(r)} - \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \bar{\mathbf{u}}^{(r)}; \mu) - \nabla_h \bar{p}^{(r),\circ} \right\rangle_{M_u}, \quad (17)$$

$$[A_r]_{ij} = \left\langle \phi_i^{u,(r)}, \mathcal{L}_h \phi_j^{u,(r)} - \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \phi_j^{u,(r)}; \mu) - \mathcal{N}_h(\phi_j^{u,(r)}, \bar{\mathbf{u}}^{(r)}; \mu) \right\rangle_{M_u}, \quad (18)$$

$$[H_r]_{ijk} = - \left\langle \phi_i^{u,(r)}, \mathcal{N}_h(\phi_j^{u,(r)}, \phi_k^{u,(r)}; \mu) \right\rangle_{M_u}, \quad (19)$$

$$[C_r^p]_{i\ell} = - \left\langle \phi_i^{u,(r)}, \nabla_h(\mu) \phi_\ell^{p,(r)} \right\rangle_{M_u}. \quad (20)$$

These terms define the backbone $\mathcal{F}_r^G(\mathbf{a}, \mathbf{b}; \mu)$ used in Eq. (2).

B.2 Gauge-consistent pressure–Poisson map

At the continuous level the gauge-fixed pressure satisfies

$$-\Delta p^\circ = \nabla \cdot [(\mathbf{u} \cdot \nabla) \mathbf{u} - \mathbf{f}], \quad \int_\Omega p^\circ \, d\Omega = 0, \quad (21)$$

with boundary conditions inherited from the momentum equation. Projection of the discrete relation gives

$$K_r^p(\mu) \mathbf{b}^{PP,(r)} = \mathbf{d}_r(\mu) + B_r(\mu) \mathbf{a}^{(r)} + Q_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}), \quad (22)$$

and hence

$$\mathcal{Q}_r^{PP}(\mathbf{a}; \mu) = [K_r^p(\mu)]^{-1} [\mathbf{d}_r(\mu) + B_r(\mu) \mathbf{a} + Q_r(\mu) : (\mathbf{a} \otimes \mathbf{a})]. \quad (23)$$

The gauge removes the constant pressure nullspace. Each complete chart therefore contains its own parameter region, affine means, bases, scalars, $\mathcal{F}_r^G$, and $\mathcal{Q}_r^{PP}$. Equal reduced dimensions do not create a canonical componentwise identification between two charts; only decoded physical fields share a common representation.

C Specialist architecture and closed-loop objective

Each regime-local chart is equipped with an independently parameterized semi-explicit specialist. The velocity coefficients evolve under the sum of the chart-local Galerkin vector field and a learned closure,

$$\dot{\mathbf{a}}^{(r)} = \mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu) + \boldsymbol{\rho}_{\theta_r}^u(\boldsymbol{\xi}^{(r)}). \quad (24)$$

Pressure is algebraic rather than an independently evolved recurrent state. After the velocity macro-step, the local pressure–Poisson operator provides the physics-based candidate

$$\mathbf{b}_{n+1}^{PP,(r)} = \mathcal{Q}_r^{PP}(\mathbf{a}_{n+1}^{(r)}; \mu), \quad (25)$$

while a data-driven candidate and a componentwise gate are evaluated from the specialist feature vector,

$$\hat{\mathbf{b}}_{n+1}^{D,(r)} = \rho_{\theta_r}^p(\boldsymbol{\xi}_n^{(r)}), \quad \boldsymbol{\alpha}_n^{(r)} = \text{sigmoid}(\boldsymbol{\eta}_{\theta_r}^p(\boldsymbol{\xi}_n^{(r)})). \quad (26)$$

The next pressure state is the componentwise convex combination

$$\mathbf{b}_{n+1}^{(r)} = \boldsymbol{\alpha}_n^{(r)} \odot \mathbf{b}_{n+1}^{PP,(r)} + (\mathbf{1} - \boldsymbol{\alpha}_n^{(r)}) \odot \hat{\mathbf{b}}_{n+1}^{D,(r)}, \quad \boldsymbol{\alpha}_n^{(r)} \in (0, 1)^{d_p}. \quad (27)$$

C.1 Features and structured sparse experts

Let

$$\mathbf{g}_n^{(r)} = \mathcal{F}_r^G(\mathbf{a}_n^{(r)}, \mathbf{b}_n^{(r)}; \mu) \quad (28)$$

be the current projected right-hand side, and let tildes denote standardization with statistics from the corresponding regime-local training population. The current and two preceding feature blocks are

$$\boldsymbol{\chi}_n^{(r)} = \text{col}(\tilde{\mathbf{a}}_n^{(r)}, \tilde{\mathbf{b}}_n^{(r)}, \tilde{\mathbf{g}}_n^{(r)}, \tilde{\mu}), \quad (29)$$

$$\mathbf{h}_{n,\ell}^{(r)} = \text{col}(\tilde{\mathbf{a}}_{n-\ell}^{(r)}, \tilde{\mathbf{b}}_{n-\ell}^{(r)}, \tilde{\mathbf{g}}_{n-\ell}^{(r)}, \tilde{\mu}), \quad \ell \in \{1, 2\}. \quad (30)$$

Thus the complete input is the three-state history

$$\boldsymbol{\xi}_n^{(r)} = \text{col}(\boldsymbol{\chi}_n^{(r)}, \mathbf{h}_{n,1}^{(r)}, \mathbf{h}_{n,2}^{(r)}). \quad (31)$$

The sampling interval is supplied to the numerical integrator; it is not an additional trainable network feature.

Within each specialist, the group router first makes a hard Top-1 decision,

$$g^* = \arg \max_{g \in \mathcal{G}} \pi_g^{\text{grp}}(\boldsymbol{\xi}). \quad (32)$$

Velocity and pressure then use separate channel routers. For $c \in \{u, p\}$, let $\mathcal{T}_2^c(\boldsymbol{\xi}; g^*)$ contain the two routed experts with the largest channel-specific scores. Together with the shared expert $e = 0$, the active output is

$$\mathcal{M}_{\theta_r}^c(\boldsymbol{\xi}) = \omega_0^c E_{0,g^*}^c(\boldsymbol{\xi}) + \sum_{e \in \mathcal{T}_2^c(\boldsymbol{\xi}; g^*)} \omega_e^c E_{e,g^*}^c(\boldsymbol{\xi}), \quad \omega_e^c \geq 0, \quad \omega_0^c + \sum_{e \in \mathcal{T}_2^c} \omega_e^c = 1. \quad (33)$$

Consequently, one group, one shared expert, and two routed experts are active for each output channel. Each expert combines nonlinear, linear/affine, and low-rank bilinear branches,

$$E_{e,g}^c(\boldsymbol{\xi}) = N_{e,g}^c(\boldsymbol{\xi}) + W_{e,g}^{c,\text{lin}} \boldsymbol{\xi} + \kappa W_{e,g}^{c,o} \left[\left(U_{e,g}^{c,\top} \boldsymbol{\xi} \right) \odot \left(V_{e,g}^{c,\top} \boldsymbol{\xi} \right) \right], \quad q_b = 4, \quad \kappa = 0.05, \quad (34)$$

where q_b is the rank of the bilinear branch. Equation (34) specifies the retained structured map without assuming undocumented hidden widths.

C.2 Frozen-context advancement and closed-loop training

During a velocity macro-step, the current pressure, the recent history, and the physical parameter form a context $\mathcal{C}_n^{(r)}$ that is held fixed across all Runge–Kutta stages. Define

$$F_r(\mathbf{a}; \mathcal{C}_n^{(r)}) = \mathcal{F}_r^G(\mathbf{a}, \mathbf{b}_n^{(r)}; \mu) + \rho_{\theta_r}^u(\boldsymbol{\xi}^{(r)}(\mathbf{a}; \mathcal{C}_n^{(r)})). \quad (35)$$

All three specialists use the classical fourth-order Runge–Kutta update

$$\begin{aligned}
\mathbf{k}_1 &= F_r(\mathbf{a}_n^{(r)}; \mathcal{C}_n^{(r)}), \\
\mathbf{k}_2 &= F_r\left(\mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{2} \mathbf{k}_1; \mathcal{C}_n^{(r)}\right), \\
\mathbf{k}_3 &= F_r\left(\mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{2} \mathbf{k}_2; \mathcal{C}_n^{(r)}\right), \\
\mathbf{k}_4 &= F_r\left(\mathbf{a}_n^{(r)} + \Delta t_r(\mu) \mathbf{k}_3; \mathcal{C}_n^{(r)}\right), \\
\mathbf{a}_{n+1}^{(r)} &= \mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).
\end{aligned} \tag{36}$$

The updated velocity is supplied to Eq. (27); the history is then shifted and the autonomous rollout continues. In particular, the retained specialist definition uses this unified RK4 update for the Steady, Hopf, and Periodic models, not a separate forward-Euler step for the Steady model.

For a closed-loop rollout of length K , the model recursively consumes its own predicted states and minimizes the regime-locally normalized coefficient error

$$\mathcal{L}_r = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{d_u} \left\| \left(\hat{\mathbf{a}}_{n+k}^{(r)} - \mathbf{a}_{n+k}^{(r)} \right) \odot \boldsymbol{\sigma}_a^{(r)} \right\|_2^2 + \frac{1}{d_p} \left\| \left(\hat{\mathbf{b}}_{n+k}^{(r)} - \mathbf{b}_{n+k}^{(r)} \right) \odot \boldsymbol{\sigma}_b^{(r)} \right\|_2^2 \right]. \tag{37}$$

The retained training specification uses progressively longer horizons $K \in \{4, 8, 12, 16\}$. The retained rollout objective is Eq. (37); we do not append an unverified single routing penalty or loss weight. The three specialists are trained and selected independently; subsequent outer-router or T2-C training does not update their local bases, scalars, reduced operators, internal routers, experts, integrators, or pressure maps.

D Outer routing and T2-C details

Parameter-only routing and admissibility. The E2 router receives only the standardized scalar parameter,

$$\tilde{\mu} = \frac{\mu - \mu_R}{\sigma_R}, \quad \boldsymbol{\pi}(\mu) = \text{softmax}(\mathcal{R}_{E2}(\tilde{\mu})) = [\pi_S, \pi_H, \pi_P]^\top. \tag{38}$$

The trajectory-level probabilities are evaluated once and remain unchanged during rollout. Adjacency is fixed by

$$A_{\text{adj}} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad \mathcal{C}_{\text{top}}(k) = \{j : A_{\text{adj},kj} = 1\}, \tag{39}$$

with ordering (S, H, P) , so direct S – P fusion is excluded. If $m_j(\mathcal{H}, \mu, K) \in \{0, 1\}$ denotes validation-supported dynamical admissibility, the masked preference is

$$\tilde{\pi}_j = \frac{\mathbf{1}_{\{j \in \mathcal{C}_{\text{top}}\}} m_j \pi_j}{\sum_{k \in \{S, H, P\}} \mathbf{1}_{\{k \in \mathcal{C}_{\text{top}}\}} m_k \pi_k + \varepsilon}. \tag{40}$$

One admissible candidate gives a Top-1 prediction; an admissible adjacent pair is passed to T2-C. No ROM prediction is asserted when every mask is zero. For a pair $(1, 2)$, define

$$\bar{\pi}_1 = \frac{\pi_1}{\pi_1 + \pi_2}, \quad \bar{\pi}_2 = 1 - \bar{\pi}_1, \quad \ell^{E2} = \log \frac{\bar{\pi}_1 + \varepsilon}{\bar{\pi}_2 + \varepsilon}. \tag{41}$$

Chart-independent physical-history descriptor. T2-C, unlike E2, uses the current physical state and two preceding states. Let $|\Omega| = \mathbf{1}^\top \mathbf{w}$, $\Delta t_n = t_n - t_{n-1}$, and let U_{ref} , P_{ref} , and

T_{ref} be fixed reference scales. With gauge-corrected pressure p_n° , define

$$\hat{E}_n^u = \frac{\|\mathbf{u}_n\|_{M_u}^2}{|\Omega|U_{\text{ref}}^2}, \quad \hat{E}_n^p = \frac{\|p_n^\circ\|_{M_p}^2}{|\Omega|P_{\text{ref}}^2}, \quad (42)$$

$$\hat{D}_n^u = \frac{T_{\text{ref}}\|\mathbf{u}_n - \mathbf{u}_{n-1}\|_{M_u}}{\Delta t_n \sqrt{|\Omega|}U_{\text{ref}}}, \quad \hat{D}_n^p = \frac{T_{\text{ref}}\|p_n^\circ - p_{n-1}^\circ\|_{M_p}}{\Delta t_n \sqrt{|\Omega|}P_{\text{ref}}}, \quad (43)$$

$$\hat{g}_n = \frac{T_{\text{ref}}}{\Delta t_n} \log \frac{\hat{E}_n^u + \varepsilon_E}{\hat{E}_{n-1}^u + \varepsilon_E}. \quad (44)$$

The second-order velocity indicator is

$$\mathbf{A}_n^u = \frac{2}{\Delta t_n + \Delta t_{n-1}} \left(\frac{\mathbf{u}_n - \mathbf{u}_{n-1}}{\Delta t_n} - \frac{\mathbf{u}_{n-1} - \mathbf{u}_{n-2}}{\Delta t_{n-1}} \right), \quad (45)$$

$$\hat{\kappa}_n^u = \frac{T_{\text{ref}}^2 \|\mathbf{A}_n^u\|_{M_u}}{\sqrt{|\Omega|}U_{\text{ref}}}. \quad (46)$$

The six statistics $\mathbf{s}_n = [\hat{E}_n^u, \hat{E}_n^p, \hat{D}_n^u, \hat{D}_n^p, \hat{g}_n, \hat{\kappa}_n^u]^\top$ are standardized as

$$\mathbf{d}_n = (\mathbf{s}_n - \boldsymbol{\mu}_s) \oslash (\boldsymbol{\sigma}_s + \varepsilon_s \mathbf{1}) \in \mathbb{R}^6. \quad (47)$$

Because these quantities are computed from physical fields, they do not depend on either candidate's POD coordinates.

Correction gate and objective. The gate input $[\tilde{\mu}; \mathbf{d}_n] \in \mathbb{R}^7$ is processed by two hidden layers of width 64 with SiLU activations and a scalar linear output. The output layer is zero-initialized, so training begins from the E2 pair probability. The gate predicts

$$\delta = c_\vartheta([\tilde{\mu}; \mathbf{d}_n]), \quad \alpha = \text{sigmoid}(\ell^{E2} + \delta). \quad (48)$$

Both specialists start from the same three-state physical history, encode it in their native charts, and perform independent autonomous rollouts. The scalar α is fixed over the prediction window. Velocity and gauge-corrected pressure are reconstructed on the common mesh and combined convexly; the fused output is never projected back into either rollout.

For candidate field errors $\mathbf{e}_1^c$ and $\mathbf{e}_2^c$, $c \in \{u, p\}$, define

$$Q^c(\alpha) = \alpha^2 \|\mathbf{e}_1^c\|_{M_c}^2 + (1 - \alpha)^2 \|\mathbf{e}_2^c\|_{M_c}^2 + 2\alpha(1 - \alpha) \langle \mathbf{e}_1^c, \mathbf{e}_2^c \rangle_{M_c}. \quad (49)$$

The correction map minimizes

$$\mathcal{L}_{\text{T2C}} = \text{mean}_{w,k} \left[\frac{Q^u(\alpha^{(w)})}{\|\mathbf{u}_{\text{true}}\|_{M_u}^2 + \varepsilon} + \frac{Q^p(\alpha^{(w)})}{\|p_{\text{true}}^\circ\|_{M_p}^2 + \varepsilon} \right]. \quad (50)$$

Gate checkpoints are selected on validation data by lexicographically minimizing the worst Reynolds-number mean error and then the global mean error. The equal blend fixes $\alpha = 1/2$; the probability blend sets $\alpha = \bar{\pi}_1$; and the convex oracle chooses its weight a posteriori for each evaluation window. The oracle is a reference, not a deployable method.

E Reproducibility and implementation details

Held-out-by-*Re* protocol and retained runs. The split unit is a complete Reynolds-number trajectory, not an individual snapshot or a randomly sampled time window. Each local specialist is fitted only on its stated training trajectories, model and routing choices use the validation trajectories, and the test trajectories are retained for the reported autonomous rollouts. A Reynolds number that occurs in two overlapping fluidic-pinball charts keeps the same split role in both. The numerical entries in this manuscript come from fixed, validation-selected runs; they are not test-selected or averaged over repeated fits.

Unified autonomous integrator. All steady, Hopf, and periodic specialists use the same classical fourth-order Runge–Kutta update with a frozen context within each macro-step. Specifically, the pressure coefficients, recent-history features, and physical parameter are held fixed across the four RK stages; only the trial velocity state changes from stage to stage. After the RK4 velocity update, the algebraic pressure closure is evaluated, the history is shifted, and the resulting predictions are fed back at the next macro-step. The rollout is initialized from the available physical history, and no reference state from inside the prediction window is injected thereafter.

Data contracts and local dimensions. For the circular cylinder, all three charts use $(r_u, r_p) = (32, 32)$. For the centered square, (r_u, r_p) is $(5, 4)$, $(11, 11)$, and $(28, 26)$ for the steady, Hopf, and periodic charts, respectively. Its 120 complete trajectories cover $Re \in [50, 150]$ on a 9,400-cell finite-volume mesh; the CFD step is $\delta t_{\text{CFD}} = 0.02$, and one snapshot is retained every 200 CFD steps, so $\Delta t_{\text{snap}} = 4$. For fluidic pinball, the corresponding ranks are $(4, 3)$, $(5, 5)$, and $(17, 16)$. Its 131 complete trajectories cover $Re \in [10, 60]$, with 94/19/18 train/validation/test trajectories overall, on a 70,194-cell mesh; $\delta t_{\text{CFD}} = 0.005$ and snapshots are retained every 0.25 physical time units. The retained circular-cylinder source uses regime-native snapshot intervals but does not provide a verified common CFD/snapshot cadence, so no single value is asserted here. Chart-wise Reynolds-number ranges, split counts, and rollout horizons are listed in Table 1.

Admissible routing and fusion. The outer regime graph permits only adjacent S – H and H – P pairs; neither a direct S – P pair nor a three-specialist route is evaluated. The parameter-only E2 probabilities are evaluated once for a complete fixed- Re trajectory and remain fixed throughout its rollout. In an admissible Top-2 case, the two specialists encode the same initial physical history, evolve independently in their native coordinates, and are combined only after reconstruction in the common physical space. Rollout windows and routing/fusion choices are fixed using validation data before the retained test evaluations.

Current evidence boundary. The retained experiment records do not specify the optimizer, learning-rate schedule, batch size, epoch counts or curriculum phase boundaries, random seeds, final hidden-layer widths, total routed group/expert counts, total parameter counts, per-query active parameter counts, or the numerical threshold used by the original divergence evaluator. These omissions prevent an exact end-to-end retraining claim. In particular, the available evidence does not justify describing the Vanilla comparison as capacity matched or parameter matched, and fixed-run differences cannot be interpreted as statistical significance without repeated seeds and dispersion estimates. A reproducibility release should add a machine-independent configuration manifest, split lists, final checkpoint identifiers, exact evaluator settings, preprocessing and basis construction, projected operators, rollout scripts, and table-generation scripts.

F Circular-cylinder results

Table 5 reports the original range over held-out long-horizon rollouts. “N/E” denotes the original not-evaluable entry.

In the S – H overlap, E2 Top-1 gives $E_u = 0.0357$ – 0.0410% and $E_p = 1.6746$ – 1.9028% . T2-C reduces these ranges to 0.0148 – 0.0355% and 0.7387 – 1.7212% , with mean steady weight spanning 0.0861 – 0.9987 . The global model yields $E_u = 0.0851$ – 0.0953% and $E_p = 1.9517$ – 2.3106% on the same overlap evaluation.

G Centered-square results

The aggregate local-specialist errors are $(E_u, E_p) = (1.5311, 1.4583)\%$ for steady flow, $(0.4387, 0.4844)\%$ for Hopf-onset flow, and $(0.9395, 1.0921)\%$ for periodic flow. On the aggregate S – H evaluation, the steady and Hopf specialists have joint errors 4.8426% and 1.7298% ; on H – P , the periodic and Hopf specialists give 1.2079% and 2.8718% . These

Table 5: Circular-cylinder long-horizon error ranges (%).

Regime	Model	E_u	E_p
Steady	RAL-MoE-ROM	0.0330–0.1723	0.2500–1.3249
	Vanilla	0.0384–0.2995	2.2163–18.5603
	Data only	0.1637–0.5122	4751.1025–27275.4645
	Global	0.0845–0.3404	1.5226–7.4363
Hopf	RAL-MoE-ROM	0.0032–0.0288	0.0355–0.4321
	Vanilla	N/E	N/E
	Data only	0.1260–0.3880	1280.2811–1524.5949
	Global	0.0110–0.0268	0.0560–0.2882
Periodic	RAL-MoE-ROM	0.3809–1.3473	1.6825–5.9601
	Vanilla	1.0393–3.0836	3.9659–13.3257
	Data only	0.7854–1.8677	2.2511–10.8189
	Global	0.5873–1.1726	2.1896–6.3566

Table 6: Parameter-wise centered-square routing and T2-C results at $K = 24$. The mean weight is the steady weight for $S-H$ and the periodic weight for $H-P$.

Overlap	Re	E2 choice	$\bar{\alpha}$	E2 joint	T2-C joint	reduction
$S-H$	95.1	Hopf	0.0967	1.7140	1.1827	31.00%
$S-H$	95.3	Hopf	0.0971	1.7456	1.2090	30.74%
$H-P$	100.5	Hopf	0.5870	2.6485	0.8571	67.64%
$H-P$	102.0	Periodic	0.7094	1.0462	0.8360	20.09%

results show why the best local choice can change between overlaps and why a corrected convex assembly is useful.

Figures 3 and 4 provide the complete velocity, gauge-corrected pressure, and pointwise-error views underlying the compact main-text comparison.

H Fluidic-pinball results

The fluidic-pinball dataset contains 131 trajectories over $Re \in [10, 60]$, with 94/19/18 train/validation/test trajectories overall. A POD-DeepONet one-step baseline gives $E_u = 1.5921\%$ and $E_p = 23.3447\%$ on 4,228 held-out input-output pairs, but only 1 of 28 held-out trajectories remains finite under recursive rollout. This contrast motivates evaluation under autonomous use rather than one-step accuracy alone.

Table 7: Fluidic-pinball overlap joint errors (%).

Overlap	K	primary	adjacent	E2-prob.	T2-C	oracle
$S-H$	24	0.6014	3.1925	1.7511	0.5897	0.5889
$S-H$	56	0.6927	3.1193	1.7714	0.6767	0.6763
$H-P$	24	0.0990	0.5146	0.2972	0.0945	0.0928
$H-P$	56	0.1028	0.5346	0.3085	0.0978	0.0955

T2-C improves the primary single-specialist prediction in all four overlap evaluations and remains close to the window-wise convex oracle. The result is consistent across the short and long rollout horizons shown here.

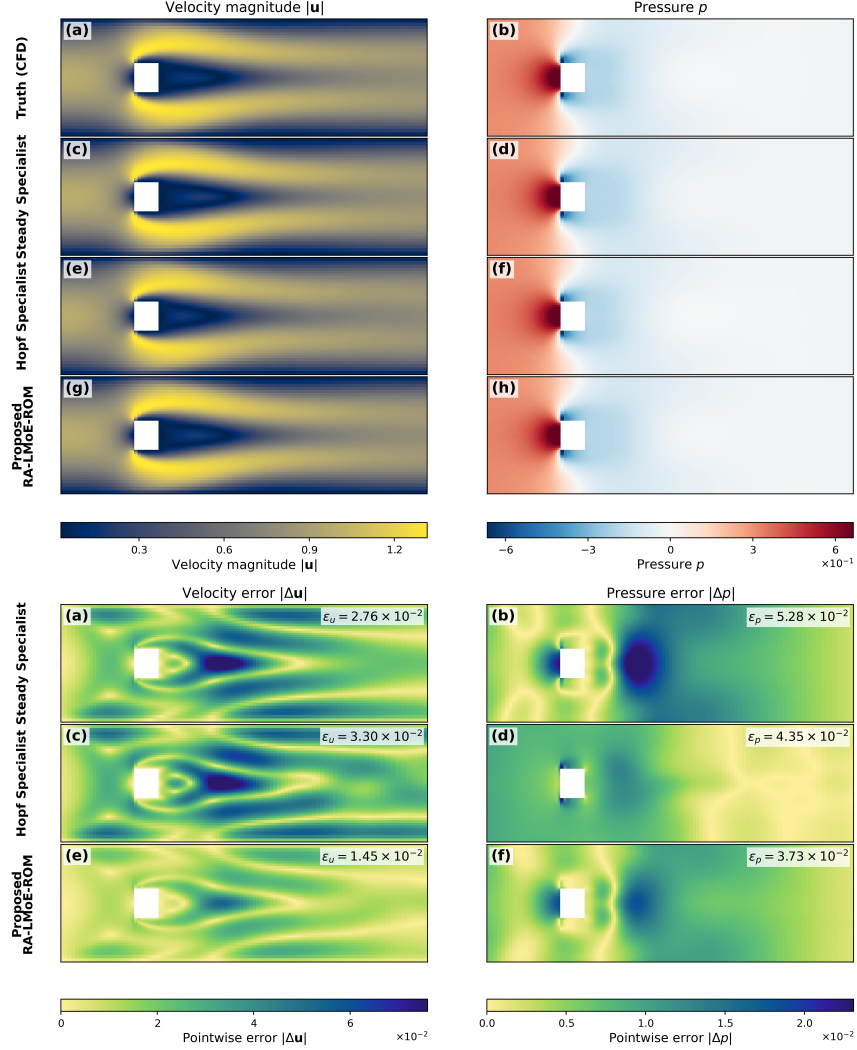


Figure 3: Representative held-out prediction in the centered-square $S-H$ overlap at $Re = 95.1$. The upper panel compares the CFD reference, Steady specialist, Hopf specialist, and T2-C prediction for velocity magnitude and gauge-corrected pressure. The lower panel gives the corresponding pointwise absolute errors.

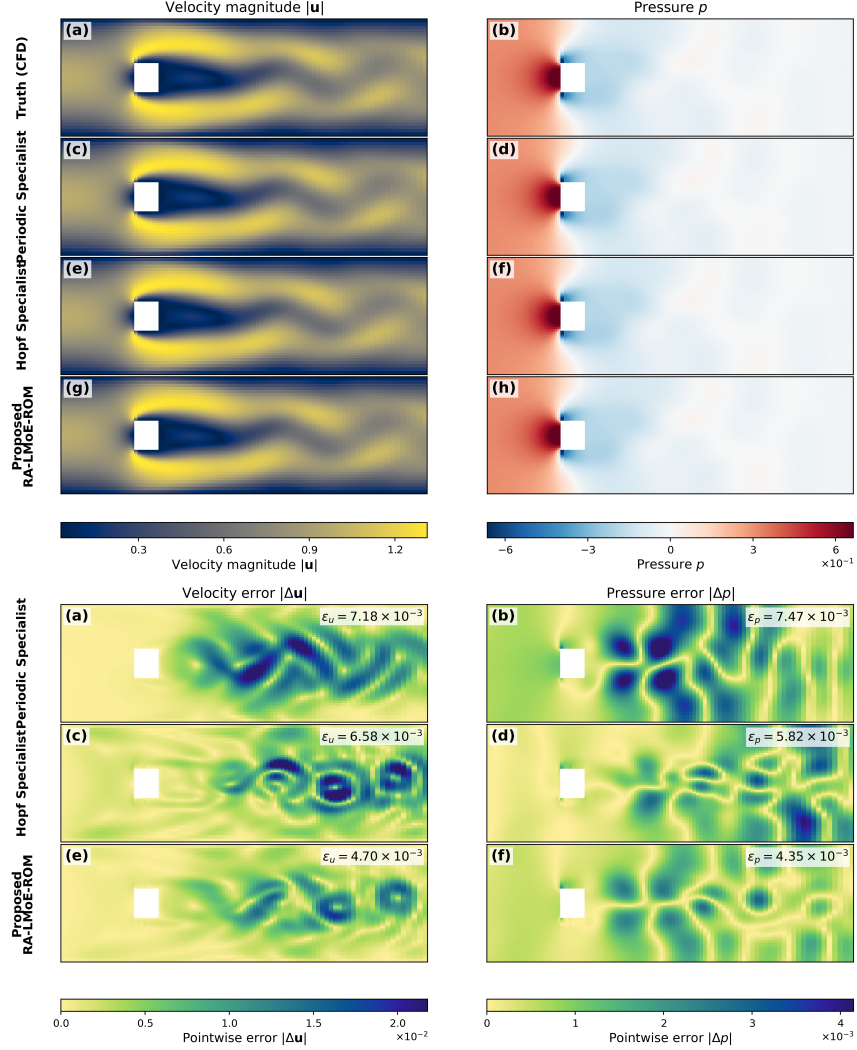


Figure 4: Representative held-out prediction in the centered-square H - P overlap at $Re = 100.5$. The upper panel compares the CFD reference, Periodic specialist, Hopf specialist, and T2-C prediction for velocity magnitude and gauge-corrected pressure. The lower panel gives the corresponding pointwise absolute errors.